

Rajasthan Board Class 12

Subject – Chemistry

Time: $3\frac{1}{4}$ Hours

M.M: 56

Section - A

1. At low temperature hydrogen is which type of molecular solid?

Answer:

At low temperature hydrogen is non-polar type of molecular solid.

2. How much part of an atom of the primitive cubic unit cell actually belongs to a particular unit cell?

Answer:

One-eighth ($1/8^{\text{th}}$) part of an atom of the primitive unit cell actually belongs to a particular unit cell.

3. Dislocation of which ion in "Ag Br" develops Frenkel defect?

Answer:

Dislocation of Ag^+ ion causes Frenkel defect in Ag Br.

4. Write the formula to calculate the mole fraction?

Answer:

The molar fraction can be represented by X. If the solution consists of components A and B, then we get

$$\text{Mole fraction of A} = \frac{\text{Moles of A}}{\text{Moles of A} + \text{Moles of B}}$$

And,

$$\text{Mole fraction of B} = \frac{\text{Moles of B}}{\text{Moles of A} + \text{Moles of B}}$$

5. What will be the value of Van't Hoff factor for ethanoic acid in benzene?

Answer:

Ethanoic acid forms dimer in benzene so Van't Hoff factor for ethanoic acid in benzene will be $1/2$.

6. What happens when a raw mango placed in concentrated salt solution?

Answer:

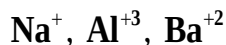
If a raw mango is placed in concentrated salt solution, it loses water due to osmosis. Water flows from solution of lower concentration to higher concentration causing raw mango to shrink.

7. Due to which forces physisorption arises?

Answer.

Physisorption arises due to the weak Van Der Waal's forces of attraction.

- 8. Arrange the following ions according to their flocculation power in the coagulation of negative sol.**



Answer:

Order of flocculation power in the coagulation of negative sol is as follows:



- 9. Why zinc is not a transition element?**

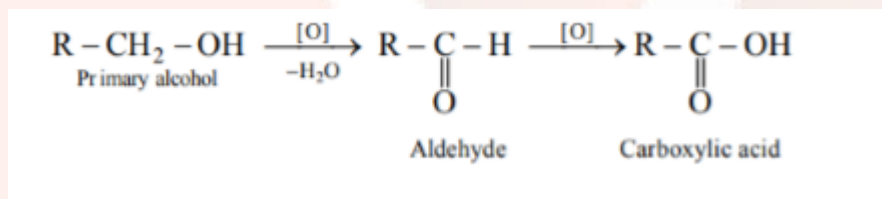
Answer:

Zn atom has completely filled penultimate d-orbitals ($3d^{10}$) in its ground state as well as in its oxidation state (Zn^{2+}), such that it is not regarded as a transition element.

- 10. Which series of compounds can be prepared by oxidation of primary alcohols?**

Answer:

Carboxyl compounds (Aldehydes and Carboxylic acids) can be prepared by oxidation of primary alcohols.



- 11. Write definition of homo polymer and co-polymer?**

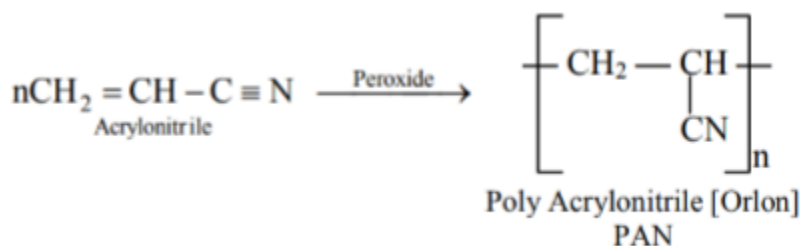
Answer:

Homo polymers is termed the addition polymers formed by the polymerization of single monomeric species like Natural rubber. While, Copolymers is termed as the addition polymers formed by the polymerization of two different monomers like Bunna-N.

- 12. Write the chemical reaction to prepare, that polymer which is used for preparation of industrial fibers by addition polymerisation of nitrile containing three carbon atoms in presence of peroxide catalyst?**

Answer:

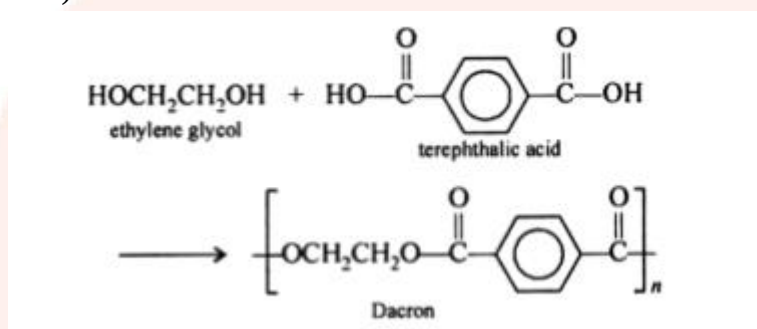
Polyacrylonitrile, is the polymer which is used for the preparation of industrial fibers by addition polymerization of nitrile containing three carbon atom is presence of peroxide catalyst.



13. Write the IUPAC suffix of that functional group of the compound which is used to prepare dacron other than ethylene glycol.

Answer:

Dacron is a polymer of ethylene glycol and terephthalic acid (benzene 1, 4 dicarboxylic acid)



Section – B

14. Resistance of a conductivity cell filled with 0.15molL^{-1} NaCl solution is 50Ω . If resistance of the same cell when filled with 0.02molL^{-1} NaCl solution is 500Ω , calculate the conductivity of 0.02molL^{-1} solution NaCl. (The conductivity of 0.15molL^{-1} NaCl solution is 1.5S/m).

Answer:

For 0.15 MNaCl solution

Resistance, $R_1 = 50\Omega$

And Conductivity, $\kappa_1 = 1.5\text{S/m}$

Now, Cell constant = $\text{Conductivity} \times \text{Resistance}$

$= 1.5 \times 50$

$= 75\text{m}^{-1}$

For 0.02 MNaCl solution

Resistance, $R_2 = 500\Omega$

Therefore,

Conductivity of 0.02 MNaCl solution,

$$\begin{aligned}\kappa_2 &= \frac{\text{Cell constant}}{\text{Resistance}} \\ &= \frac{75}{500} \\ &= 0.15 \text{ s/m}\end{aligned}$$

15. The conductivity of 0.011 mol L^{-1} acetic acid is $5 \times 10^{-5} \text{ cm}^{-1}$. Calculate its dissociation constant if λ_k^0 for acetic acid is $250 \text{ Scm}^2 \text{ mol}^{-1}$?

Answer:

We have,

Concentration of acetic acid is 0.011 mol L^{-1} , conductivity $5 \times 10^{-5} \text{ cm}^{-1}$

$$\lambda_m^0 = 250 \text{ Scm}^2 \text{ mol}^{-1}$$

Now,

$$\lambda_m = \frac{k}{c}$$

$$\Rightarrow \lambda_M = \frac{5 \times 10^{-5} \text{ cm}^{-1}}{0.001}$$

$$\Rightarrow \lambda_M = 50 \text{ Scm}^2 \text{ mol}^{-1}$$

Degree of dissociation

$$\alpha = \frac{\lambda_m}{\lambda_m^0} = \frac{50}{250} = 0.2$$

Therefore,

Dissociation constant

$$K_\alpha = \frac{c\alpha^2}{1-\alpha}$$

$$\Rightarrow K_\alpha = \frac{0.00001(0.2)^2}{1-0.2}$$

$$\Rightarrow K_\alpha = \frac{4 \times 10^{-5}}{0.8}$$

$$\Rightarrow K_\alpha = 5 \times 10^{-5}$$

16. . Explain adsorption from solution phase? Write Freundlich equation related to it?

Answer.

The following observations have been made in the case of adsorption from solution phase:

- (i) The extent of adsorption decreases with increase in temperature.
- (ii) The extent of adsorption increases with increase in surface area.

- (iii) Amount of adsorption is directly proportional to concentration of the solute in solution.
- (iv) It also depends on the nature of the adsorbent and the adsorbate.

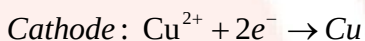
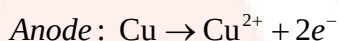
$$\frac{x}{m} = kC^{1/n} \quad [\because n > 1]$$

Where, x is the amount of adsorbate in gm, m is the amount adsorbent in gm, c is the concentration of solute, k and n are the constants which depend on the nature of adsorbate and adsorbent.

17. Explain copper is refined with electrolytic method? Explain with necessary reaction?

Answer:

An acidified solution of copper sulfate is taken as electrolyte. The net result of electrolysis is the transfer of copper in pure form from the anode to the cathode.



18. Explain colour and complex formation tendency of transition metal ions?

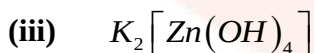
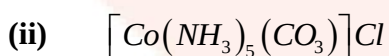
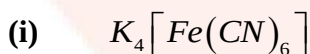
Answer:

The transition metal compounds are colored both in the solid state and in solution. And the colour of the transition metal ion is due to the d- d transition. The amount of energy required to excite some of the electrons to higher energy states within the same d-sub shell corresponds to the energy of certain colour of visible light.

Transition metals have strong tendency to form complex ions and complex compounds. It is due to the following reasons

- Comparatively smaller size of the metal ions.
- High ionic charges.
- Availability of d orbitals for bond formation.

19. Classify the following into homoleptic and heteroleptic complexes?



Answer:

Homoleptic are those complexes in which a metal is bound to only one kind of donor groups. While heteroleptic are those complexes in which a metal is bound to more than one kind of donor group.

$K_4[Fe(CN)_6]$ And $K_2[Zn(OH)_4]$ are heteroleptic

$[Co(NH_3)_5(CO_3)]Cl$ And $[Pt(NH_3)_2Cl(NO_2)]$ are homoleptic

20. Explain the violet colour of $[Ti(H_2O)_6]^{3+}$ complex on basis of the crystal field theory ?

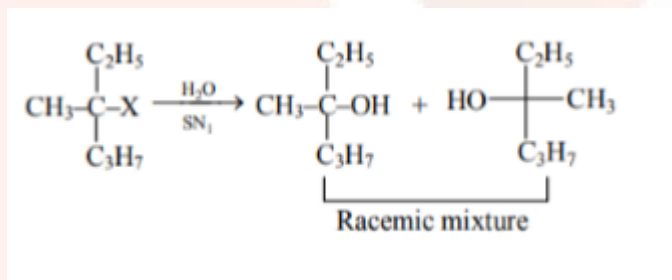
Answer.

We know that in ground state Titanium has 23 electrons with outer shell $3d^3 4s^2$. In $[Ti(H_2O)_6]^{3+}$ such that we get titanium has +3 oxidation state. Single electron in the metal d orbital is in t_{2g} level in the ground state of the complex. The next higher state available for the electron is the empty e_g level.

21. . Explain SN_1 reaction with appropriate example.

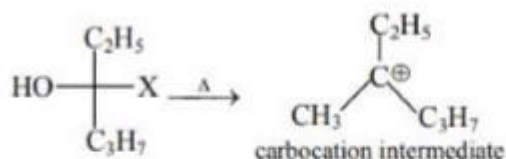
Answer.

It is the nucleophilic substitution in which the rate of reaction depends on concentration of only one component like SN_1 . It is unimolecular, proceeding through an intermediate carbocation.

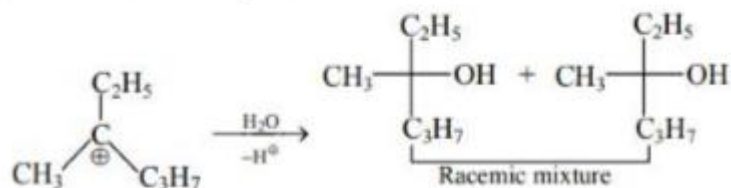


And its mechanism is:

(A) Formation of carbocation



(B) Attack of Nucleophile

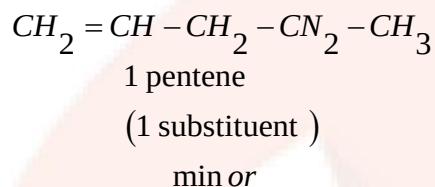
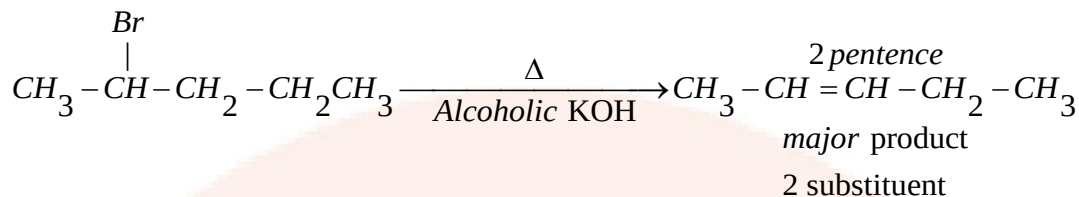


OR

What is Saytzeff (Zaitser) rule? Explain elimination reaction in 2-bromopentane.

Answer.

Saytzeff (Zaitser) rule states that in dehydrohalogenation reactions, the preferred product is that alkene which has the greater number of alkyl groups attached to the doubly bonded carbon atoms.



22. Explain why aryl halides are less reactive towards nucleophilic substitution reactions?

Answer.

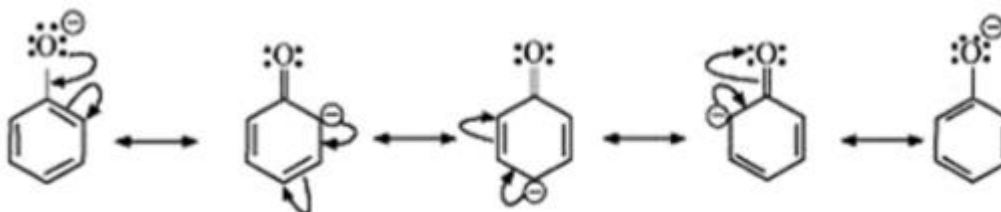
The reasons for less reactive nature of aryl halides are as follows:

- Resonance effect as in aryl halides there is a conjugation of electron pairs of halogen atom with pi electrons of the benzene ring. And the electrons attached on C-X bond.
- The sp^2 hybridised carbon atom involved in C-X bond in haloarene molecule is more electronegative than the sp^3 hybrid carbon atom in alkyl halide so the sp^2 hybridised carbon has less tendency to release electrons to the halogen atom.

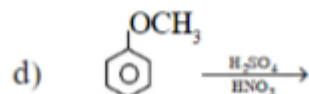
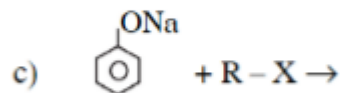
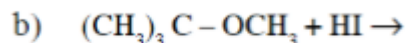
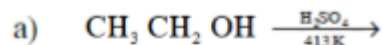
23. Explain why hydroxyl group attached to an aromatic ring is more acidic than the one in which hydroxyl group is attached to an alkyl group?

Answer:

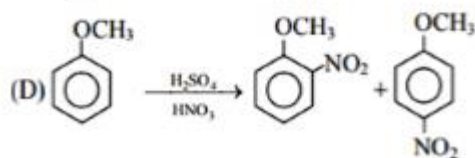
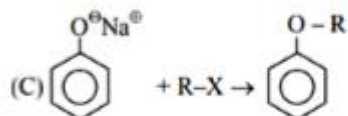
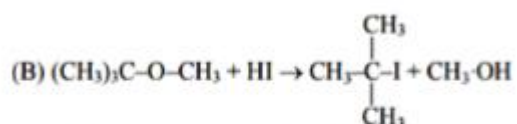
In first case the greater acidity because the stability of the phenoxide ion, which is resonance.



24. Complete the following chemical reactions:



Answer:



Section – C

25. (a) What do you mean by amonolysis of alkyl halides give chemical reaction?

(b) Write that reaction which is used for following :

(i) Ascent of amine series.

(ii) To synthesis amine containing one carbon less than that of in the amide.

Answer:

(a) Alkyl halides on reaction with ammonia yield primary, secondary and tertiary amines as given below:

26. (a) What are monosaccharides? Write the definition?

(b) (i) Name the products yields from the complete hydrolysis of DNA?

(ii) Write the structural formula of sugar moiety in DNA & RNA?

(c) Write the name of method related with DNA to identify the dead body in any accident?

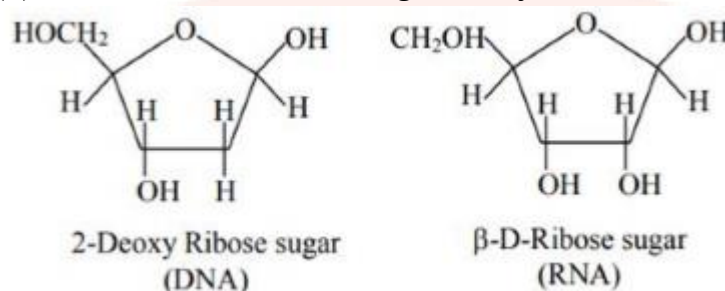
Answer.

(a) Carbohydrates which cannot be further hydrolysed are named as monosaccharides. For example, glucose, fructose, galactose, etc.

(b) (i) Product of DNA hydrolysis are as follows:

- Pentose sugar (β -2 deoxy ribose sugar)
- Nitrogenous base (Adenine, Guanine, Thymine, Cytosine)
- Phosphoric acid

(ii) Structural formula of sugar moiety in DNA & RNA are:



(c) DNA fingerprinting is the method used to identify the dead body in any accident.

27. (a) What are tranquilizers? Write the name of any mild tranquilizer?

(b) How food preservatives work. Give an example of food preservative?

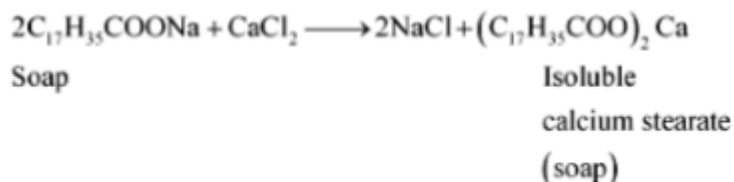
(c) Why do soaps not work in hard water?

Answer:

(a) Tranquilizer is the drug which acts on the central nervous system and helps in reducing anxiety and mental tension like Chlorodiazepoxide.

(b) Food preservatives help to protect food materials against microorganisms by killing or inhibition of bacteria like, sodium benzoate.

(c) Hard water contains calcium and magnesium ions. Such that these ions displace sodium or potassium from their salts and form insoluble calcium or magnesium salts of fatty acids.



OR

27. (a) How drugs can be classified according to different criteria. Write the name of each?

(b) How antihistamines work? Give an example of any antihistamine?

(c) Write the structural formula of any sleep-producing barbiturate?

Answer:

(a) Drugs can be classified on the following basis:

- (i) Pharmacological effect
- (ii) Biochemical process
- (iii) Chemical structure
- (iv) Molecular targets

(b) Antihistamines are defined as the drugs which combat the effects of histamine a chemical released by certain cells of the body.

The commonly used antihistamines are as follows:

- Diphenyl hydramine (Benadryl)
- Chlopheniramine (chlortrimeton)
- Fexofenadine (Allegra)
- Cetirizine (Zyrtec)

Section – D

28. (a) What are the factors which effect the rate of a chemical reaction?

(b) Derive the rate constant formula for zero order reaction?

(c) Draw a diagram showing effect of catalyst on activation energy.

Answer.

(a) Factors affecting the rate of chemical reaction are

- (i) Nature and concentration of reactants
- (ii) Activation energy (E_a)
- (iii) Temperature
- (iv) Presence of catalyst

(b) A reaction is said to be of zero order if its rate is independent of the concentration of the reactants.

Let $R \rightarrow P$ is zero order reaction.

$$\therefore \text{Rate} = \frac{-d[R]}{dt} = k[R]^0$$

$$\Rightarrow \text{Rate} = \frac{-d[R]}{dt} = k$$

$$\Rightarrow d[R] = -kdt$$

By integrating both sides,

$$[R] = -kt + C \dots (1)$$

Where, C is the constant of integration.

$t = 0$ The concentration of the reactant $R = [R]_0$, where $[R]_0$ is initial concentration of the reactant.

$$[R]_0 = -k \times 0 + c$$

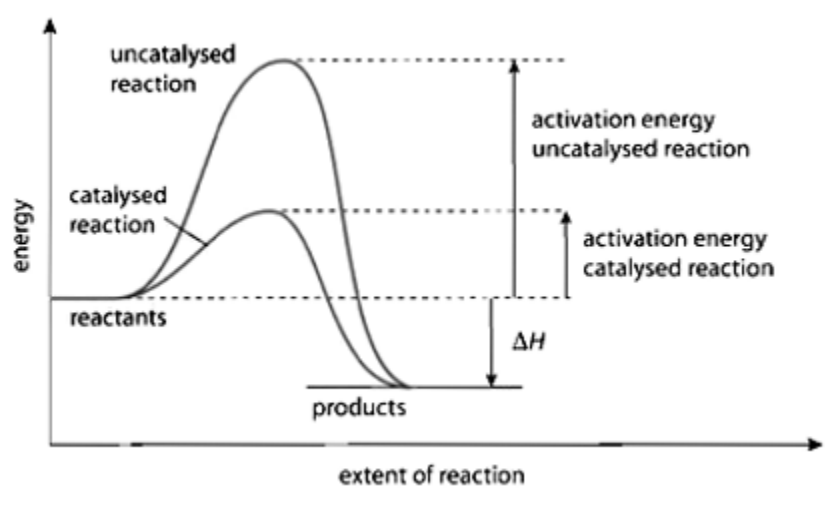
$$C = [R]_0$$

Now, put the value of C in eq (i)

$$[R] = -kt + [R]_0$$

$$\Rightarrow k = \frac{[R]_0 - [R]}{t}$$

(c) The diagram showing effect of catalyst on activation energy of a reaction:



OR

28. (a) What is pseudo first order reaction? Inversion of cane sugar is which type of reaction. Write its chemical reaction?

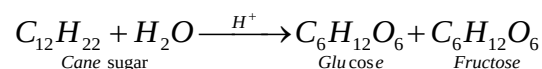
(b) Calculate the half-life of the first order reaction whose rate constant is $5 \times 10^{-14} \text{ s}^{-1}$.

(c) Plot a graph between $\ln k$ and $1/T$ on the basis of Arrhenius equation.

Answer:

(a) Pseudo order reaction states that reaction whose actual order is different from that expected by using rate law expression.

Inversion of cane sugar is pseudo I order type of reaction.



(b) Half-life of the first order reaction is calculated as follows

$$t_{\frac{1}{2}} = \frac{0.693}{k}$$

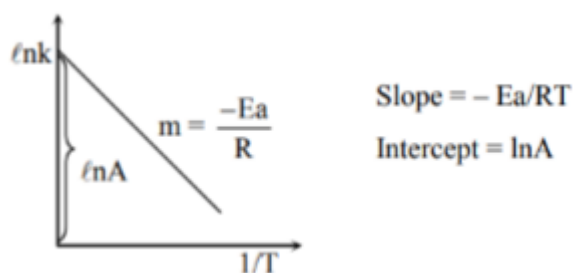
$$\Rightarrow t_{\frac{1}{2}} = \frac{0.693}{5 \times 10^{-14}}$$

$$\Rightarrow t_{\frac{1}{2}} = 0.13 \times 10^{14} \text{ s}$$

$$k = Ae^{-E_a/RT}$$

$$\ln k = \ln A - \frac{E_a}{RT}$$

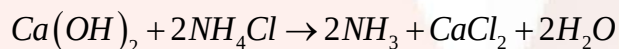
Comparing above equation with equation of straight line, $y = c + mx$.



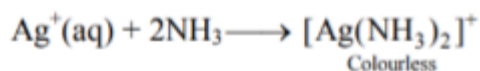
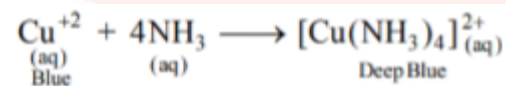
29. (a) Write a chemical equation to prepare ammonia from ammonium chloride? (b) Write a chemical equation to identify Cu^{2+} and Ag^{+} ions with the application of NH_3 .

(c) Draw a labeled diagram showing flowchart for the manufacture of ammonia?
Answer:

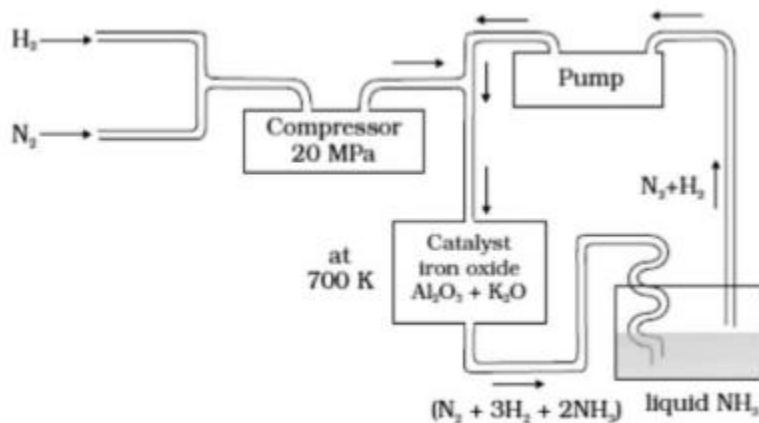
(a) On small-scale ammonia is obtained from ammonium chloride which decomposes when treated with caustic soda (lime).



(b) presence of a lone pair of electrons on the nitrogen atom of the ammonia molecule makes it a Lewis base which reacts with Cu^{2+} and Ag^{2+} ions to give two differently coloured complexes are given below:



(c) Flowchart for the manufacture of ammonia is given below:



OR

29. (a) Write the chemical equation to prepare sulphur dioxide from industrial method?

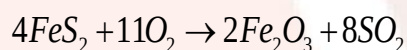
(b) What happens when SO_2 is passed through water?

(c) Write reaction showing reducing property of sulphur dioxide?

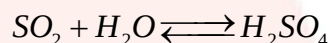
(d) Draw a labeled diagram showing ring structure of rhombic sulphur?

Answer.

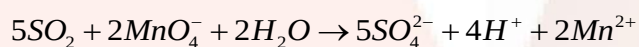
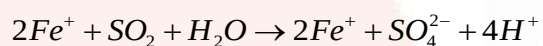
(a) Sulphur dioxide is prepared by roasting of sulphide ore:



(b) When SO_2 is passed through water, sulphurous acid is formed



(c) Reactions showing the reducing property of SO_2 are as follows:



(d) Ring structure of rhombic sulphur, S_8 is:

